

TRAIN ACCIDENTS, FIRES & DERAILMENTS

Revision Guide — Driver Rules & Procedures

GERT8000-M1 Iss 9 | GERT8000-M3 Iss 5 | EMR PPTs 25-01, 25-03, 26-03

1. WHAT COUNTS AS A TRAIN ACCIDENT?

The rule book (M1 §1) defines a train accident as any of the following:

- A **derailment**
- A **collision** involving trains or rail vehicles
- A collision with an **obstruction**
- A collision with a **road vehicle**
- A collision with a **person**
- A **fire** on a train that might put other trains in danger, or that requires passenger evacuation onto running lines
- An **accidental train division** that has caused another line to be obstructed

Source: GERT8000-M1 §1

Why Accidents Are Investigated

Legal requirements under ROGS 2006 (Railways and Other Guided Transport Systems Safety Regulations). Corporate manslaughter legislation also applies.

Purpose: establish events leading up to the accident, determine immediate and underlying causes, and consider the role of infrastructure and train-borne systems. (EMR PPT 25-01 slides 6–7)

2. IMMEDIATE ACTIONS AFTER A TRAIN ACCIDENT

Source: GERT8000-M1 §2.1; EMR PPT 25-01 slide 8; EMR PPT 26-03 slide 10

Step 1 — Hazard Warning (Do This FIRST)

Switch on the **hazard warning indication** immediately.
If you cannot — display a **red light forward** instead.

Step 2 — Contact the Signaller (REC)

Tell the signaller about the accident and whether the **traction current needs to be switched off**.

Contact in this order of preference:

1. Make a **Railway Emergency Call (REC)** from your driving cab.
2. Any other direct communication from the cab if REC is not possible.
3. Make a REC from another driving cab on the train.

If you have to leave your cab to use another cab, place a **TCOC on any other obstructed line FIRST**.

Step 3 — Stay in the Cab

Once you have contacted the signaller, **stay in the cab** until the signaller confirms:

- Every driver whose train must remain at a stand has received and understood the message, OR
- No train is approaching.

Step 4 — If the Signaller Says a Train Is Approaching

Leave your train IMMEDIATELY.

Proceed **TOWARDS** the approaching train showing a **hand danger signal**.

2A. Structure of the Emergency Call (REC)

State in this order:

- "This is an emergency call"
- Who you are and your **train reporting number**
- **Where** you are speaking from
- **What** has happened
- The **exact location** of the train accident
- **Which lines** are obstructed
- Whether the **traction current** needs to be switched off
- Whether the **emergency services** are needed

Source: EMR PPT 26-03 slide 13; M1 §2.1

2B. If You Cannot Contact the Signaller from the Train

You must leave the train to contact the signaller. Before leaving:

- Find out which lines are obstructed (if in doubt, **treat them as obstructed**).
- Place a **TCOC on each obstructed line**.
- **Tell the guard** (if provided) that you are leaving the train.

Proceed towards the line more likely to have an approaching train, or where you will reach the signaller more quickly. Show a hand danger signal.

Contact the signaller via: any direct communication outside the train → lineside telephone → signal box.

Source: M1 §2.1

3. LEAVING THE TRAIN — 4 T'S AND AN S

When walking to the signaller to raise the alarm, place **3 detonators 20 metres apart** and display a hand danger signal if you reach any of the following before the full protection distance:

4 T's and an S — Place 3 Detonators When You Reach:

- T — A **Train approaches** (place detonators immediately and show hand danger signal)
- T — A **Tunnel entrance** — place detonators here; proceed through only if: tunnel is short, communication likely on the other side, you can walk safely and show a hand danger signal. Otherwise stay at the entrance.
- T — A **Trailing/diverging junction** — place detonators, stay at the junction showing hand danger signal to approaching trains.
- T — A **Telephone** or lineside phone — place detonators, then contact the signaller.
- S — A **Signal box** (Absolute Block type) — place detonators, then speak with the signaller.

If you stop an approaching train: tell that driver what has happened and ask them to contact the signaller by REC.

Source: GERT8000-M1 §3; EMR PPT 26-03 slide 17 ("4 Teas and a Sugar")

Multi-Track Lines

Place emergency protection on EVERY line that is obstructed.
This may require placing protection ahead AND behind your train.
The guard or other competent person can assist.
If the full protection distance falls within a tunnel, continue through to the other side and place protection there.
(EMR PPT 26-03 slide 14, 18)

3A. Protecting Your Own Line

After protecting other obstructed lines, you may be required to protect the line your train is standing on if:

- Extended Block Working (EBW) is in operation, OR
- Your train has been **completely derailed** (place TCOC on your own line when signaller tells you to).

Source: M1 §2.1; EMR PPT 26-03 slide 16, 23

4. FIRE ON A TRAIN

Source: GERT8000-M1 §4; GERT8000-G1; EMR PPT 25-03

DROP → STOP → BOX (Do in This Order)

DROP — Drop the pantograph / shut down the diesel engine (unless needed to reach safety) / switch off ETS. This can be done on the move. Any electrical ventilation will also switch off, limiting fire spread. Voltage remaining on train will be 110V — AFFF extinguishers can then be used safely.

STOP — Stop in a suitable location. AVOID: tunnels, viaducts, or other unsuitable places. Suitable locations include: near a level crossing, next to a road, at a station (unless canopy is wooden), or close to a station.

BOX — Contact the signaller using Emergency Call (REC) on the GSM-R cab radio as first preference.

Any fire, however small, on a moving train can spread very quickly and must be extinguished promptly and completely. If the fire cannot be put out within a few seconds, stop the train immediately.

Source: M1 §4.1

4A. After Stopping — Assess and Act

- Advise passengers using the PA to move to an **unaffected coach**.
- Assess the situation — if a larger fire, **request emergency services** (summon all if unsure who to call).
- If the fire will affect other trains and/or passengers are already evacuating — switch on **hazard/red tail lamps**, then **protect other running lines first**.
- If the fire affects the OLE — arrange for **OLE isolation**.
- If other lines are NOT affected — deal with the incident and evacuate passengers if necessary (lines must be blocked and protected first).
- **Lateral evacuation** is the first choice for evacuating passengers.
- If there is a danger of fire spreading and it is safe to do so — **separate the burning vehicle** from the rest of the train.

Source: M1 §4.2–4.3; EMR PPT 25-03 slide 9

Always Protect Running Lines BEFORE Dealing With Passengers

You must protect all other affected running lines FIRST, before dealing with your passengers.
(M1 §4; EMR PPT 25-03 slide 10)

4B. Reporting a Fire — Order of Preference

Priority	Method
1st	Emergency Call button on GSM-R cab radio
2nd	Signal Post Telephone (SPT)
3rd	Lineside phone
4th	Mobile phone
5th	Any other available means

Always use the Emergency Call procedure even for a small fire — they can get out of hand very quickly.

Source: EMR PPT 25-03 slide 8

4C. When the Fire Is Out

Situation	Action
Fire out — train CANNOT proceed	Tell the signaller. Carry out any necessary protection. Passengers left on train and safe: arrange controlled evacuation when possible. Passengers who have left the train: ensure they are in a safe position

	away from electrified lines and running lines — keep them there until arrangements are made to escort them.
Fire out — train CAN proceed	Tell the signaller as soon as possible. (M1 §4.5)

Source: M1 §4.4–4.5

4D. Dangerous Goods

If Conveying Dangerous Goods

Alert the controlling signaller and state:

"THIS IS A RAIL DANGEROUS GOODS EMERGENCY"

Then deal with the incident as per a dangerous goods incident. (EMR PPT 25-03 slide 13)

4E. Fire Extinguisher Classes

Class / Type	Best Extinguisher
Class A — Freely burning (cloth, wood, paper, fabric)	Water or spray foam
Class B — Flammable liquids (oils, spirits, petrol)	Foam, spray foam, or dry powder
Class C — Flammable gases (propane, butane, gas)	Dry powder only
Electrical — live electrical equipment	Isolate power first; if power cannot be isolated use CO2 or dry powder (non-conductive)

Source: EMR PPT 25-03 slides 4–5

4F. Using an Extinguisher Safely

- Only tackle a fire if it is **SAFE** to do so.
- Always keep yourself **between the fire and the escape route**.
- Do not attempt to fight a fire in a confined space.
- Before tackling electrical equipment — **isolate the supply immediately**.
- Check: tamper seals and safety pins removed; discharge hoses are not cut; no obstructions in nozzle.

Source: EMR PPT 25-03 slide 3

5. SEEING SOMETHING WRONG WITH ANOTHER TRAIN

If you see any of the following, contact the signaller immediately:

- A door not closed properly
- An insecure load on a freight train
- A vehicle on fire
- A hot axle box
- The headlight not lit
- The tail lamp missing or not lit
- The driver sounding the **train-in-distress warning** (continuous series of long blasts on the high/loud tone of the horn)
- The driver or guard displaying a **red handsignal**
- The **hazard warning indicator** (flashing headlights)

Source: EMR PPT 25-03 slide 14

5A. Hazards That Endanger Other Trains

The following require you to stop approaching trains:

- A track defect
- A flood
- An obstruction
- A fire
- Any light which is out at an emergency indicator
- A **cow, bull, or other large animal** within the boundary fence (even if not immediately dangerous to trains)
- Any other animals on or near the line

Source: EMR PPT 25-03 slide 15

6. TRAIN DERAILMENTS

Source: GERT8000-M1 §2.1; GERT8000-M3 §2; EMR PPT 26-03

After a derailment, follow the same immediate accident actions (Section 2): hazard warning → REC to signaller → check obstructed lines → TCOC on other obstructed lines.

Fully Derailed — Place TCOC on Your Own Line

If the whole train has been derailed, or might not be operating track circuits, you must ALSO place a TCOC on the line your train is standing on when the signaller tells you to.
(M1 §2.1; EMR PPT 26-03 slide 11)

6A. After the Derailment — What Can Run?

Vehicle / Situation	Rule
Derailed vehicle	Must NOT continue in service until examined by a **Rolling Stock Technician (RST)**.
Vehicle coupled NEXT TO the derailed vehicle	Must also be examined by an RST. Leave coupled until examined if possible.
Unaffected portion of the derailed train	Must also be examined before continuing in service.
Derailment at LOW SPEED	Vehicles that were not derailed, or coupled next to a derailed vehicle, may be examined at the **first suitable location**. (M3 §2.1)

Source: M3 §2.1; EMR PPT 26-03 slide 18

6B. Evacuating After a Derailment

Controlled Evacuation Steps (EMR PPT 26-03 slide 19)

Make arrangements for a controlled evacuation.
Communicate with person in charge of movements to arrange the evacuation.
Communicate with on-train staff and passengers.
Egress driving compartment prior to commencing evacuation.
Arrange for local opening of doors to facilitate controlled evacuation.
Facilitate passengers' safe egress from the train.
Move passengers to a position of safety.
⚠ REMEMBER: Before ANY evacuation — LINES MUST BE BLOCKED AND PROTECTED FIRST.

7. EVACUATING A TRAIN

Source: GERT8000-M1 §6

Only Evacuate if ABSOLUTELY Necessary

Lines MUST be blocked and protected first.

BEFORE any evacuation, the signaller must have stopped trains on all lines that may be affected. (M1 §6.1)

Type	When	What to Do
Controlled evacuation	Planned — conditions allow time to coordinate	Tell the signaller you are evacuating and ask them to stop all approaching trains on affected lines. Ask for traction current off if needed. Wait for signaller to confirm no train approaching, then tell the guard. (M1 §6.3)
Emergency evacuation	Immediate danger — no time to wait	Carry out Section 2.1 actions. Tell the signaller an emergency evacuation is taking place. Ask for traction current off if needed. If the signaller says a train is approaching — proceed on foot TOWARDS it showing a hand danger signal. (M1 §6.4)
Uncontrolled evacuation	Passengers have already left or are leaving without instruction	Carry out Section 2.1 actions. Tell the signaller an uncontrolled evacuation is taking place. Ask for traction current off if needed. Try to prevent passengers making an uncontrolled evacuation and warn those who have evacuated about the risks. (M1 §6.6)

After an evacuation, ensure passengers who have left the train are in a safe position away from electrified lines and other running lines. Keep them there until arrangements are made to escort them from the site.

8. PRESERVING EVIDENCE

Source: GERT8000-M3 §1; EMR PPT 25-01 slide 9

Emergency action always takes priority over preserving evidence. Your priorities in order after a serious accident:

Priority	Action
1	Secure the train
2	Protect the line
3	Get traction current switched off
4	Call the emergency services
5	Make sure the public and other personnel are safe

Source: M3 §1.1

8A. Entering the Driving Cab After an Accident

You must only enter a driving cab if you have to:

- Secure the train
- Carry out rescue operations
- Use the cab radio to contact emergency services or the signaller
- Get equipment to protect the line or extinguish a fire
- Make a note of **short-life evidence** (e.g. wheel temperatures, brake-gauge readings)

If You Enter the Cab

Be careful NOT to disturb equipment, handles, buttons, or switches unnecessarily. Afterwards, make a detailed note of the original position of all equipment and gauges. Give this to an investigating officer BEFORE you leave the scene. (M3 §1.3)

8B. Evidence — What Not to Touch

- Do NOT disturb or interfere with evidence unless told to by: a **Rail Incident Officer (RIO)**, a **police incident officer**, a **RAIB investigator**, or any other senior investigator.
- Make an immediate note of evidence likely to be lost — wheel/tyre temperatures, brake-block temperatures, brake-gauge readings.

Source: M3 §1.2

8C. After the Incident

- The emergency services take charge of serious incidents, setting up cordons.
- Network Rail sends a **Rail Incident Officer (RIO)** to site.
- The TOC/FOC sends a **Train Operating Liaison Officer (TOLO)**.
- A **routine drugs and alcohol test** will be arranged.
- An initial interview will take place to establish the facts.
- You will be asked to complete an **incident report form**.
- **Media enquiries** must be directed to the rail media relations manager, RIO, or TOLO — not answered by you.
- Prior to leaving the scene, give all information you have recorded to the RIO, police incident officer, or senior investigator.

Source: EMR PPT 25-01 slides 2–3, 8, 16

8D. When to Complete a Driver's Report Form

- Following an **operational incident or accident**
- To explain an **operating problem**
- To explain your **actions** (e.g. late on duty)
- To explain **late running** of your train
- After **witnessing** an incident, accident, or incorrect working practices
- When **requested** to do so by your line manager

Tip: make notes in a notebook or on your schedule card at the time — this makes completing the report form much easier and more accurate.

Source: EMR PPT 25-01 slide 12

9. ACCIDENTAL TRAIN DIVISION

Source: GERT8000-M1 §5

9A. Immediate Actions

- Make sure both portions are **secure** and all vehicles are accounted for.
- Tell the guard (if provided) about the situation.
- Check couplings where the train divided — look for track/lineside damage and whether **recoupling is possible**.

9B. If the Two Portions Can Be Recoupled

- Get the **personal authority of the signaller** for the recoupling movement.
- Once recoupled, tell the signaller the train is again complete.

Source: M1 §5.3

9C. If the Two Portions Cannot Be Recoupled

Before leaving the rear portion behind, ensure:

- A **red light** is displayed at the **rear** of the portion being left behind.
- A **white light** is displayed at the **front** of the portion being left behind.

Then tell the signaller: that the rear portion is to be left in the section, and its exact location.

- Do NOT pass the next stop signal or block marker until you have told the signaller.
- Do NOT leave a single-line section until you have told the signaller.
- Fit a tail lamp on the rear of the front portion — on a TCB line, fit it immediately; on other lines, fit it when the front portion reaches the next signal box or a TCB line.

Source: M1 §5.4

Assisting Train — Speed Limit

The signaller will tell the assisting train driver the exact location of the divided portion and the colour of the light displayed on it.

Proceed at CAUTION. Do not exceed **25 mph (40 km/h)**. (M1 §5.5)

10. QUICK REFERENCE — KEY POINTS

Accident Immediate Priority Order

1. Hazard warning on (or red light forward)
2. REC to signaller — state: emergency call / who you are / train number / location / what happened / lines obstructed / traction current needed? / emergency services needed?
3. Stay in cab until signaller confirms protection provided or no train approaching
4. If train approaching — leave train IMMEDIATELY and walk towards it with hand danger signal
5. If cannot contact signaller from train — place TCOC on obstructed lines, tell guard, walk to telephone/signal box

Before Any Evacuation

Lines MUST be blocked and protected FIRST.
Only evacuate if absolutely necessary. (M1 §6.1)

Fire — DROP STOP BOX

DROP: pantograph / diesel / ETS
STOP: suitable location — NOT a tunnel or viaduct
BOX: contact signaller by REC
Protect running lines BEFORE dealing with passengers.

4 T's and an S — When to Place 3 Detonators

T — Train approaches
T — Tunnel entrance
T — Trailing/diverging junction
T — Telephone
S — Signal box (Absolute Block type)
Place 3 detonators 20 metres apart each time.

Key Numbers

25 mph (40 km/h) — max speed of assisting train removing rear portion of divided train (M1 §5.5)
20 metres — spacing between each of 3 detonators
3 detonators — placed at each of the 4T + S situations

Situation	Rule
Derailed vehicle	Must not run until examined by Rolling Stock Technician (RST)
Vehicle coupled next to derailed vehicle	Also needs RST examination; leave coupled until examined if possible
Derailment at slow speed	Non-derailed vehicles may be examined at first suitable location
Collision or heavy impact	Must not continue until examined by RST (M3 §2.2)
Entering cab after accident	Only for: securing train, rescue, cab radio use, get protection equipment, note short-life evidence (M3 §1.3)
Media enquiries	Direct to RIO, TOLO, or media relations manager — not you